



In Search of the Computational Primitives of Language

Aniello De Santo

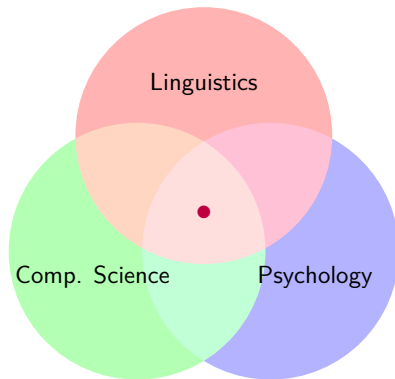
he/him

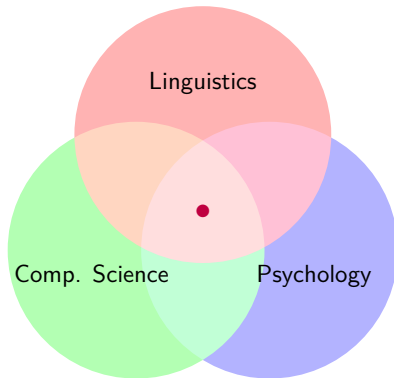
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Linguistic Primitives

- ▶ What are the core representations?
- ▶ What are the core computational operations?
- ▶ What do they tell us about processing/learning mechanisms?

Computation and Theory Building

*[...] this is a confusion of two quite separate issues, **simulation and explanation**. [...] What we are **really** interested in [...] is explanation — in developing models that help us **understand how it is that people behave** that way, not merely demonstrating that we can build an artifact that behaves similarly.*

(Kaplan, 1995)

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Interpretability for the win!

A Lens: Formal Language Theory

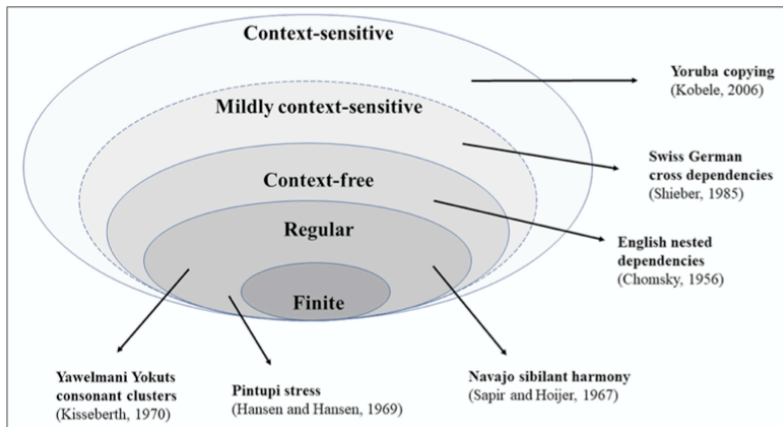
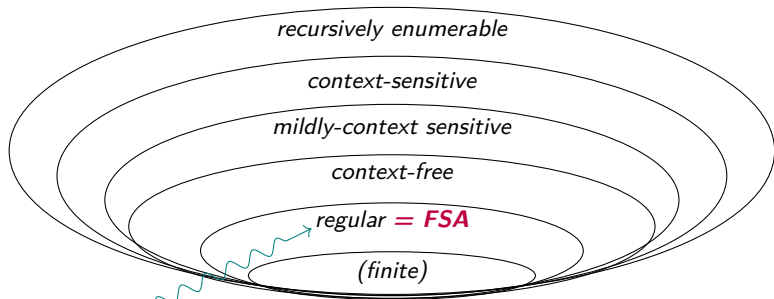


Figure 1: The Chomsky Hierarchy. Various features of natural language occupy different regions of the hierarchy. Figure reproduced from Figure 1 in Heinz (2010: 634) with permission.

Spoken Languages' Phonology as a Regular System



Phonology

Kaplan and Kay (1994)

Local Phonotactic Dependencies

1 Intervocalic voicing in Italian

Forbid voiceless segments in between two vowels

- (1) a. * /kasa/
b. /kaza/
→ cf. orthography: “*casa*”

Intervocalic voicing is Strictly Local (SL)

- ▶ Forbid voiceless segments in-between two vowels: *V[-voice]V
- ▶ **Italian:** *ase, *ise, *ese, *isi, ...

\$ k a s a \$

\$ k a z a \$

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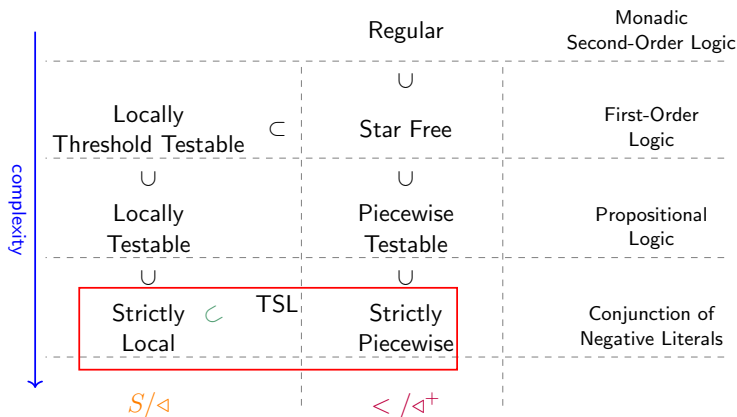
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Beyond Automata: Subregular Languages¹



¹McNaughton & Papert (1976), Heinz (2011), Chandlee & Heinz (2014), De Santo & Graf (2019), De Santo & Rawski (2022), a.o.

FLT, Linguistics, and LLM Expressivity

Saturated Transformers are Constant-Depth Threshold Circuits

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Between Circuits and Chomsky: Pre-pretraining on Formal Languages Imparts Linguistic Biases

Michael Y. Hu¹ Jackson Petty² Chuan Shi¹ William Merrill¹ Tal Linzen^{1,2}

Neuro-Symbolic Language Modeling with Automaton-augmented Retrieval

Uri Alon¹ Frank F. Xu¹ Junxian He¹ Sudipta Sengupta² Dan Roth³ Graham Neubig¹
¹Language Technologies Institute, Carnegie Mellon University ²Amazon AWS ³AWS AI Labs
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What Makes Instruction Learning Hard? An Investigation and a New Challenge in a Synthetic Environment

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Switching Gears: Benchmarking with Sentence Processing

Not All Structures Are Processed Equally

- Subject VS object relative clause

SRC **The horse** [_{RC} that **t** kicked the wolf] went home.

ORC **The horse** [_{RC} that the wolf kicked **t**] went home.

- Attachment preferences

1.a I shot an elephant in my pajamas

1.b I [shot an elephant] [in my pajamas]

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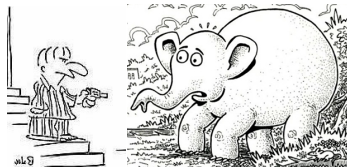
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Assessing the Ability of LSTMs to Learn Syntax-Sensitive Dependencies

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Ambiguity All the Way Down



mcsnuggie

true self control is waiting until the movie starts to eat your popcorn



leftforbed

why would the movie eat my popcorn



leftforbed

nevermind i get it

Ambiguity All the Way Down



Ambiguity All the Way Down

Ambiguity is ubiquitous in natural language!

Why Care? The Cognitive Science answer

- ▶ How do humans handle multiple structural representations?
- ▶ What principles guide ambiguity resolution cross-linguistically?
- ▶ Language specific properties vs. general biases/mechanisms?

Why Care? The NLP answer

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Ambiguity and Relative Clauses Cross-linguistically

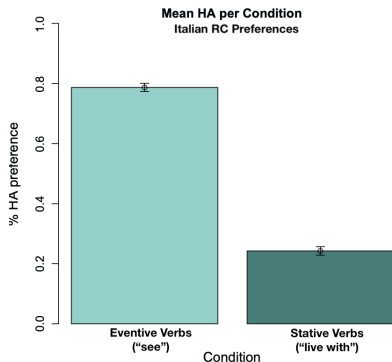
- ▶ They saw the daughter of the actress that was on the balcony

NP₁ **The daughter** was on the balcony HA

NP₂ **The actress** was on the balcony LA

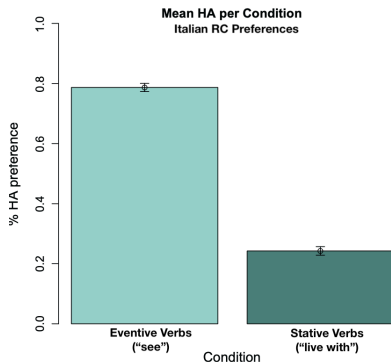
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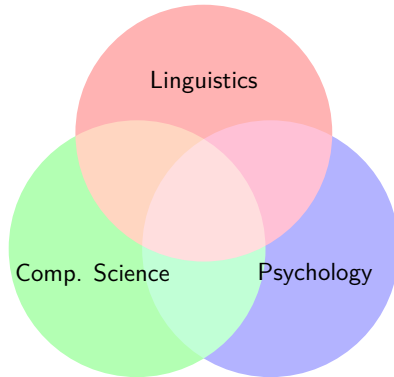
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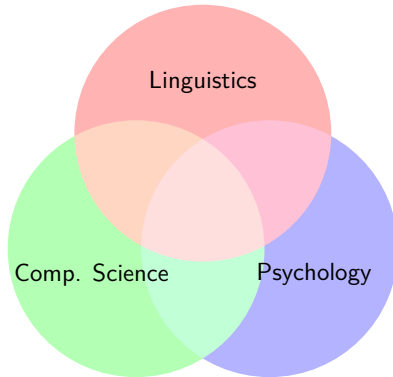


LLMs and Italian RCs?



Check out the poster!





Thank you!